

Nekotiator: LLM-based Opponent Modeling

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1 Introduction

In automated negotiation, an agent’s ability to reach a mutually beneficial agreement heavily depends on its capacity to understand the opponent’s preferences. Traditional opponent models often rely on frequency-based heuristics, observing which values the opponent proposes most often. However, these methods struggle to capture the nuances of human-like or complex opponent strategies.

To address this, we present the opponent modeling framework of Nekotiator. By integrating a Large Language Model (LLM) into the negotiation pipeline, the agent actively parses opponent messages and proposals to estimate hidden utility structures, such as issue weights and specific target values.

2 LLM-based Opponent Modeling

The core contribution of Nekotiator is its real-time opponent modeling mechanism, which directly utilizes an LLM to translate opponent behavior into mathematical utility estimations.

2.1 Preference Extraction via LLM

Upon receiving a proposal or message, the agent queries the LLM, parsing the JSON response to extract the weight and target value for each issue. These attributes are accumulated and the weights normalized to robustly estimate the opponent’s preferences.

2.2 Utility Estimation (Zero-Sum vs. Non-Zero-Sum)

To evaluate how much the opponent will appreciate a specific proposal, the agent classifies each issue into two categories: zero-sum and non-zero-sum. The estimated opponent utility for a given value x , denoted as $U_{opp}(x)$, is calculated differently based on this classification:

Zero-Sum Issues: For issues with strictly opposing interests (e.g., price), the agent assumes the opponent prefers extreme values. The normalized value score is directly mapped to the utility, assuming monotonic preference.

Non-Zero-Sum Issues: For non-zero-sum issues, the opponent’s utility is modeled based on proximity to the LLM-inferred target value T . The agent employs a distance-based peak function:

$$U_{opp}(x) = 1 - \frac{|x - T|}{D_{max}} \quad (1)$$

where D_{max} is the maximum possible deviation within the domain of the issue. This allows the agent to accurately evaluate how close a potential offer is to the opponent’s actual desires.

3 Utility-driven Negotiation Strategy

The inferred opponent model is integrated into a utility-driven framework to generate counter-proposals and make acceptance decisions.

3.1 Concession and Acceptance Criteria

Nekotiator manages its own utility through an internal aspiration level. At the beginning of the negotiation, the agent demands its maximum utility. Over time, this aspiration level gradually decays from 0.7 to 0.5. If the agent observes the opponent making concessions, it dynamically accelerates its own concession rate.

An opponent’s offer is accepted if it meets either of the following criteria:

1. The offer’s utility is greater than or equal to the agent’s current aspiration level.
2. The offer is superior to the agent’s planned next counter-proposal.

3.2 Win-Win Proposal Generation

If an offer is rejected, Nekotiator generates a counter-proposal. The agent first filters the outcome space to identify candidate offers that satisfy its current aspiration level. From this subset, it selects the candidate that maximizes the estimated opponent utility, $U_{opp}(x)$. This mechanism ensures that the agent rigorously protects its own baseline while strategically offering the most attractive compromise to the opponent.

4 Conclusion

This paper outlined the core opponent modeling architecture of Nekotiator. By leveraging LLMs to extract explicit targets and issue weights, and dynamically applying these to zero-sum and non-zero-sum evaluations, the agent transcends traditional frequency-based modeling. Coupled with a strict aspiration-based concession strategy, this approach effectively facilitates the generation of win-win agreements in complex negotiation environments.